

1. How do you create a DataFrame from a dictionary?

```
import pandas as pd
data = {'name': ['A', 'B'], 'age': [2, 3]}
df = pd.DataFrame(data)
```

2. How to check the shape, size, and data types of a DataFrame?

```
df.shape      # Returns (rows, columns)
df.size       # Total number of elements
df.dtypes     # Data types of columns
df.info()     # Summary of DataFrame
df.memory_usage(deep=True) # Memory usage
```

3. How do you get the first and last 5 rows of a DataFrame?

```
df.head()     # First 5 rows
df.tail()     # Last 5 rows
```

4. How to rename columns in a DataFrame?

```
df.rename(columns={'old_name': 'new_name'}, inplace=True)
```

5. How to reset and set the index of a DataFrame?

```
df.reset_index(drop=True, inplace=True)
df.set_index('column_name', inplace=True)
```

6. How to detect and count missing values?

```
df.isnull().sum()
```

7. How to fill missing values with mean/median/mode?

```
df['col'].fillna(df['col'].mean(), inplace=True)    # Fill with mean
df['col'].fillna(df['col'].median(), inplace=True)  # Fill with median
df['col'].fillna(df['col'].mode()[0], inplace=True) # Fill with mode
```

8. How to drop rows or columns with missing values?

```
df.dropna(axis=0, inplace=True) # Drop rows
df.dropna(axis=1, inplace=True) # Drop columns
```

9. How to detect and remove duplicates?

```
df[df.duplicated()]          # Detect duplicates
df.drop_duplicates(inplace=True) # Remove duplicates
```

10. How to replace values in a DataFrame?

```
df.replace({'old': 'new'}, inplace=True)
```

11. How to filter rows based on a condition?

```
df[df['age'] > 30]
```

12. How to filter rows using multiple conditions?

```
df[(df['age'] > 30) & (df['gender'] == 'Male')]
```

13. How to query rows using `query()`?

```
df.query("age > 30 and gender == 'Male'")
```

14. How to use `isin()` to filter values?

```
df[df['country'].isin(['India', 'USA'])]
```

15. How to apply a custom function row-wise?

```
df.apply(lambda row: row['a'] + row['b'], axis=1)
```

16. How to perform multiple aggregations?

```
df.groupby('region').agg({'sales': ['sum', 'mean']})
```

17. How to get group size and count?

```
df.groupby('category').size()  
df.groupby('category')['item'].count()
```

18. How to apply transformations to groups?

```
df.groupby('region')['sales'].transform('mean')
```

19. How to rank values within groups?

```
df['rank'] = df.groupby('region')['sales'].rank(ascending=False)
```

20. How to merge two DataFrames (like SQL JOIN)?

```
pd.merge(df1, df2, on='id', how='left')
```

21. How to concatenate DataFrames?

```
pd.concat([df1, df2], axis=0) # vertical  
pd.concat([df1, df2], axis=1) # horizontal
```

22. How to pivot data?

```
df.pivot_table(values='sales', index='region', columns='month',  
aggfunc='sum')
```

23. How to unpivot (melt) data?

```
pd.melt(df, id_vars=['id'], value_vars=['score1', 'score2'])
```

24. How to join based on index?

```
df1.join(df2, how='inner')
```

25. How to convert a column to datetime?

```
df['date'] = pd.to_datetime(df['date'])
```

26. How to extract year, month, day from a date?

```
df['year'] = df['date'].dt.year  
df['month'] = df['date'].dt.month  
df['day'] = df['date'].dt.day
```

27. How to filter rows based on date range?

```
df[(df['date'] >= '2023-01-01') & (df['date'] <= '2023-12-31')]
```

28. How to create a new column for day of week?

```
df['day_of_week'] = df['date'].dt.day_name()
```

29. How to set datetime column as index?

```
df.set_index('date', inplace=True)
```

30. How to create new columns based on other columns?

```
df['total'] = df['price'] * df['quantity']
```

31. How to use `np.where()` for conditional columns?

```
import numpy as np
df['grade'] = np.where(df['score'] > 90, 'A', 'B')
```

32. How to apply `map()` or `replace()` for value mapping?

```
df['gender'] = df['gender'].map({'M': 'Male', 'F': 'Female'})
```

33. How to apply string methods to a column?

```
df['name'] = df['name'].str.lower()
```

34. How to split a column into multiple columns?

```
df[['first', 'last']] = df['full_name'].str.split(' ', expand=True)
```

35. How to calculate correlation between features?

```
df.corr()
```

36. How to calculate cumulative sum and product?

```
df['cumsum'] = df['sales'].cumsum()
df['cumprod'] = df['returns'].cumprod()
```

37. How to calculate rolling mean?

```
df['rolling_avg'] = df['sales'].rolling(window=7).mean()
```

38. How to use `diff()` and `pct_change()`?

```
df['diff'] = df['sales'].diff()  
df['pct_change'] = df['sales'].pct_change()
```

39. How to detect outliers using IQR?

```
Q1 = df['value'].quantile(0.25)  
Q3 = df['value'].quantile(0.75)  
IQR = Q3 - Q1  
outliers = df[(df['value'] < Q1 - 1.5*IQR) | (df['value'] > Q3 +  
1.5*IQR)]
```

40. How to get summary statistics for numeric columns?

```
df.describe()
```

41. How to get value counts for categorical column?

```
df['category'].value_counts()
```

42. How to find unique values and their count?

```
df['column'].unique()  
df['column'].nunique()
```

43. How to identify skewness and kurtosis?

```
df['column'].skew()  
df['column'].kurt()
```

44. How to plot histogram and boxplot?

```
df['sales'].hist()  
df.boxplot(column='sales')
```

45. How to create a bar plot?

```
df['category'].value_counts().plot(kind='bar')
```

46. How to plot a time series?

```
df.set_index('date')['sales'].plot()
```

47. How to use Seaborn for correlation heatmap?

```
import seaborn as sns  
sns.heatmap(df.corr(), annot=True)
```

48. How to use Matplotlib for multiple plots?

```
import matplotlib.pyplot as plt  
plt.figure(figsize=(10,5))  
plt.plot(df['date'], df['sales'])  
plt.show()
```